

Stress Transfer and Energy Distribution in Functionally Graded Tendon–Bone Interfaces

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Abstract

The tendon–bone interface (enthesis) is a biologically optimized structure characterized by a gradual transition in mechanical properties between compliant tendon and stiff bone. This transition plays a crucial role in mitigating stress concentrations and ensuring effective load transfer.

In this study, a two-dimensional finite element model of the enthesis was developed in Abaqus/CAE to investigate the influence of different material transition laws on stress distribution and elastic energy storage. A plane stress formulation with CPS4R elements was adopted to isolate the effect of the material gradient under static uniaxial loading. Linear, exponential, and power-law gradients of the elastic modulus were compared against a sharp interface configuration.

Results show that graded material transitions significantly reduce stress concentrations and promote a more distributed load transfer. Among the tested configurations, the power-law model provides the best balance between peak stress reduction and spatial regularity of the stress field. A sensitivity analysis on the power-law exponent, evaluated through criteria of peak stress reduction and distribution regularity, identifies $n = 3$ as the optimal configuration.

Further analysis demonstrates that the distribution of strain energy density is strongly correlated with stress gradients and is primarily governed by the stiffness variation, while the effect of Poisson’s ratio variation is negligible.

These findings confirm the effectiveness of functionally graded materials in improving mechanical performance at biomaterial interfaces and provide insights for the design of biomimetic implantable devices.

Keywords: Functionally graded materials, enthesis, finite element method,

1. Introduction

The interface between dissimilar materials represents a critical region in many engineering and biological systems. Abrupt changes in material properties are known to generate stress concentrations, which can compromise structural integrity and lead to failure [2, 8].

In biological systems, this issue is frequently addressed through the presence of functionally graded interfaces, where material properties vary gradually across a transition region. The tendon–bone interface, known as the enthesis, is a paradigmatic example of this strategy [1, 6].

The enthesis exhibits a progressive transition in mechanical properties from the compliant tendon to the stiff bone, allowing for efficient load transfer and minimizing stress concentrations [2, 7]. This graded architecture has inspired the development of functionally graded materials (FGMs) for engineering applications, particularly in the design of biomimetic implantable devices where controlled stress transfer at dissimilar material interfaces is a primary design requirement [4].

Despite extensive experimental and theoretical studies, the mechanical role of the specific form of the material gradient remains not fully understood. In particular, the influence of different transition laws on both stress distribution and energy storage requires further investigation [3].

The aim of this study is to analyze how different material grading strategies affect the mechanical response of the tendon–bone interface. A two-dimensional finite element approach was selected, as it allows systematic parametric variation of the material gradient while maintaining computational tractability and clear interpretability of results; three-dimensional and nonlinear extensions are discussed as future work. The model compares sharp and graded configurations, including linear, exponential, and power-law transitions of the elastic modulus.

In addition to stress analysis, the distribution of strain energy density is evaluated to provide a more comprehensive understanding of the mechanical behavior of the interface.

The study follows a structured multi-phase approach, including mesh convergence assessment, sensitivity analysis on the gradient exponent, and evaluation of Poisson’s ratio effects, with the objective of identifying the

most mechanically effective material transition law. This work contributes to the understanding of graded biological interfaces by systematically linking the form of the material gradient to stress transfer mechanisms and energy distribution.

2. Materials and Methods

2.1. Geometry and Model Setup

A two-dimensional finite element model of the tendon–bone interface (enthesis) was developed in Abaqus/CAE. The geometry consisted of a rectangular domain with total length 30 mm and height 6 mm.

The domain was partitioned into three longitudinal regions:

- tendon (12 mm)
- enthesis (6 mm)
- bone (12 mm)

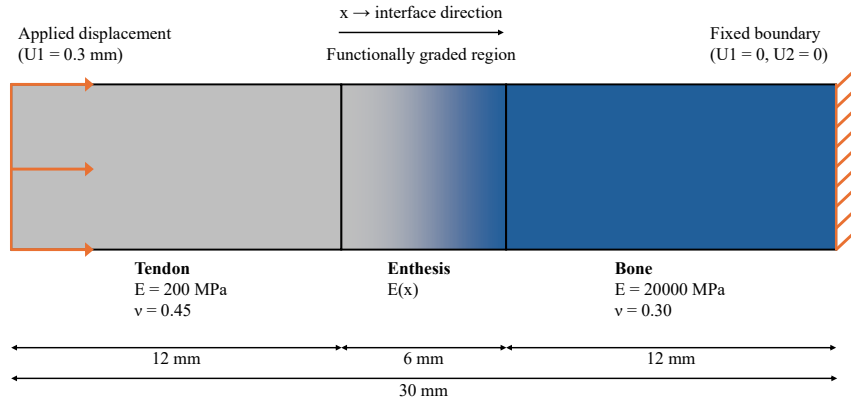


Figure 1: Schematic representation of the tendon–bone interface model. The enthesis is modeled as a functionally graded region with spatially varying elastic properties. A prescribed displacement ($U_1 = 0.3$ mm) is applied on the tendon side, while the bone side is fully constrained.

The model was implemented as a single 2D deformable part with longitudinal partitions. The enthesis region was discretized into multiple layers to approximate a continuous variation of material properties.

2.2. Material Models

All materials were modeled as linear elastic and isotropic.

The tendon was assigned a Young's modulus $E_t = 200$ MPa and Poisson's ratio $\nu_t = 0.45$, representing its nearly incompressible behavior. The bone was assigned $E_b = 20,000$ MPa and $\nu_b = 0.30$.

This results in a stiffness mismatch of approximately two orders of magnitude across the interface.

2.2.1. Elastic modulus transition laws

A local coordinate x was defined along the enthesis region, with $x = 0$ at the tendon side and $x = L_e$ at the bone side, where $L_e = 6$ mm.

The elastic modulus in tendon and bone regions was defined as:

$$E_t = 200 \text{ MPa}, \quad E_b = 20,000 \text{ MPa}$$

Different transition laws were adopted to describe the spatial variation of $E(x)$ within the enthesis.

Sharp interface. The elastic modulus is defined as a discontinuous function:

$$E(x) = \begin{cases} E_t, & x < x_i \\ E_b, & x \geq x_i \end{cases}$$

where x_i corresponds to the tendon–bone interface.

Linear gradient.

$$E(x) = E_t + (E_b - E_t) \frac{x}{L_e}$$

Exponential gradient.

$$E(x) = E_t \left(\frac{E_b}{E_t} \right)^{x/L_e}$$

Power-law gradient.

$$E(x) = E_t + (E_b - E_t) \left(\frac{x}{L_e} \right)^n$$

The exponent n controls the shape of the stiffness transition: lower values produce a smoother gradient, while higher values concentrate stiffening toward the bone side, resulting in a delayed but sharper transition.

Layer-wise discretization. The continuous functions $E(x)$ were approximated through a layer-wise discretization. Each layer was assigned the elastic modulus evaluated at its centroid, ensuring a consistent numerical representation of the graded profile.

2.3. Finite Element Implementation

A plane stress formulation was adopted, consistent with the thin geometry and uniaxial loading condition. CPS4R elements (4-node quadrilateral elements with reduced integration) were used.

A static general step was defined with geometric nonlinearity disabled (NLGEOM = OFF), allowing isolation of material effects on stress distribution.

Boundary conditions were applied as follows:

- right boundary (bone side): $U_1 = 0$, $U_2 = 0$
- left boundary (tendon side): prescribed displacement $U_1 = 0.3$ mm, U_2 free

This configuration generates a predominantly uniaxial tensile stress state along the longitudinal direction.

A structured mesh was employed, with local refinement in the enthesis region (element size ≈ 0.2 – 0.25 mm) and coarser discretization elsewhere, ensuring accurate resolution of stress gradients while maintaining computational efficiency.

2.4. Parametric Study and Model Selection

A multi-stage parametric analysis was conducted to identify the most mechanically effective graded configuration.

Phase A — Discretization refinement

The number of layers in the enthesis region (8, 12, 16) was varied to assess convergence of the discretized gradient. Convergence was evaluated based on peak S11 values and smoothness of the stress profile. The 16-layer configuration was selected for subsequent analyses.

Phase B — Power-law exponent sensitivity

A sensitivity analysis was performed on the power-law exponent ($n = 1.5, 2, 3, 5$) using 16 layers.

The evaluation criteria included:

- reduction of peak stress magnitude
- smoothness and symmetry of stress distribution
- continuity of load transfer across the interface

The configuration with $n = 3$ was identified as the optimal compromise between stress reduction and distribution regularity.

Phase C — Influence of Poisson’s ratio

Two variants of the optimal configuration were investigated:

- M17: constant Poisson’s ratio in the enthesis
- M18: spatially varying Poisson’s ratio

2.4.1. Poisson’s ratio distribution

In M17, a constant value $\nu = 0.35$ was assigned across the enthesis.

In M18, Poisson’s ratio varied from tendon to bone:

$$\nu_t = 0.45, \quad \nu_b = 0.30$$

The distribution was defined as:

$$\nu(x) = \nu_t + (\nu_b - \nu_t) \left(\frac{x}{L_e} \right)^n$$

with $n = 3$, ensuring consistency with the elastic modulus gradient.

Phase D — Energy-based validation

The strain energy density (SENER) was analyzed to evaluate the global mechanical response and the spatial distribution of stored elastic energy.

2.5. Summary of Investigated Configurations

Table 1: Summary of model configurations considered in the study.

Model	Transition type	Layers	Notes
Sharp interface	Discontinuous	—	Baseline reference
Linear gradient	Linear	8	Uniform stiffness increase
Exponential gradient	Exponential	8	Rapid stiffening near bone
Power-law	Power-law ($n = 1.5-5$)	16	Sensitivity analysis
M17	Power-law ($n = 3$)	16	Constant ν
M18	Power-law ($n = 3$)	16	Variable ν

2.6. Data Extraction and Post-Processing

Stress and energy fields were extracted along the central axis of the model (mid-height, $y \approx 3$ mm) using custom Python scripts interfacing with Abaqus output databases (.odb).

The scripts:

- extracted S11 and SENER values at integration points
- identified global extrema
- exported spatial profiles in CSV format

The analysis focused on:

- longitudinal stress (S11) distribution and peak values
- spatial gradients of S11
- strain energy density (SENER) distribution
- correlation between stress and energy fields

3. Results

3.1. Stress Distribution (S11) — Phase 1

The longitudinal stress distribution (S11) along the central axis is reported in Figure 2 for the different transition laws considered in Phase 1.

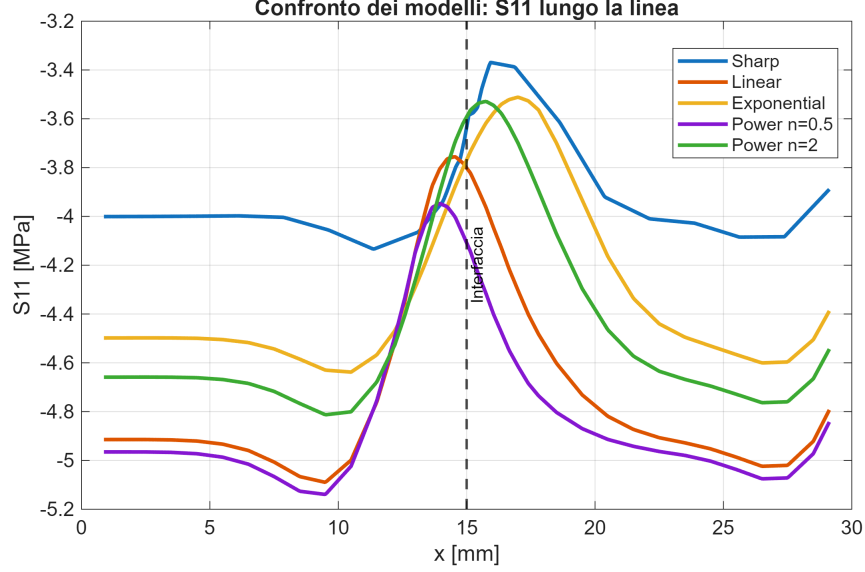


Figure 2: Longitudinal stress distribution (S_{11}) along the central axis for different material transition laws (Phase 1). The dashed line indicates the nominal tendon–bone interface at $x = 15$ mm.

The sharp interface exhibits a pronounced stress peak localized at the interface, reflecting the abrupt stiffness mismatch.

All graded configurations reduce the peak magnitude and redistribute the stress over the enthesis region. The linear and exponential models produce partial smoothing but introduce asymmetry: the linear gradient shifts the peak toward the tendon side, whereas the exponential gradient biases the response toward the bone side.

Among the tested configurations, the power-law model with $n = 2$ provides the most balanced stress distribution, characterized by reduced peak magnitude and improved spatial symmetry.

3.2. Stress Gradient Analysis

The spatial derivative of S_{11} is shown in Figure 3.

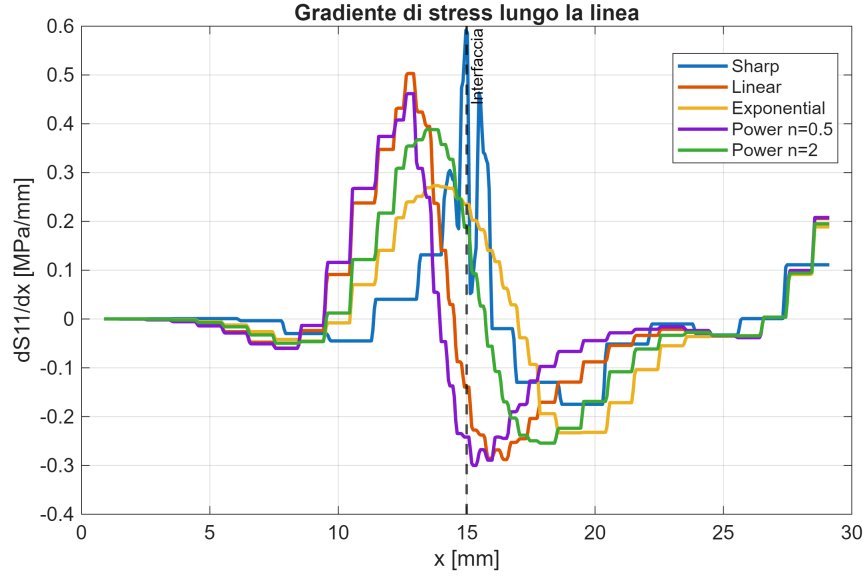


Figure 3: Spatial gradient of S11 along the central axis. Graded configurations distribute load transfer over a wider region compared to the sharp interface.

The sharp interface is associated with a highly localized gradient peak, indicating abrupt load transfer. In contrast, graded configurations spread the gradient over a wider region, reducing its intensity.

Power-law distributions exhibit the most symmetric gradient profiles, whereas linear and exponential models retain noticeable asymmetry. This behavior supports the adoption of the power-law formulation for further analysis.

3.3. Stress Curvature Analysis

The second spatial derivative of S11, representing stress curvature, is reported in Figure 4.

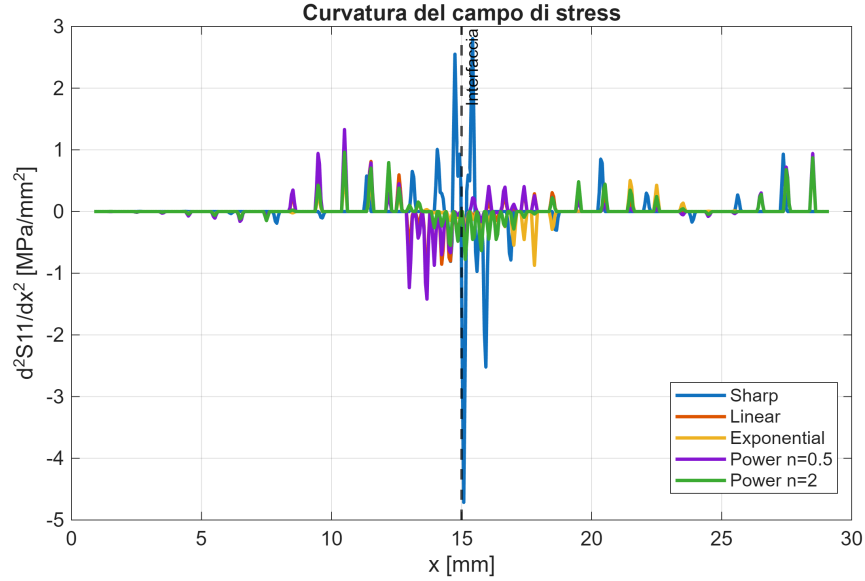


Figure 4: Second spatial derivative of S11 along the central axis. Graded models reduce curvature peaks and broaden their spatial extent.

The sharp interface produces strong, localized curvature peaks at the interface. All graded configurations reduce both the magnitude and localization of these peaks.

The power-law model with $n = 2$ exhibits the most regular curvature profile, with minimal oscillations. Lower exponents (e.g., $n = 0.5$) introduce local irregularities, indicating reduced stability of the stress field.

3.4. Quantitative Comparison — Phase 2

Table 2: Qualitative comparison of S11 characteristics for Phase 1 models.

Model	Peak S11	Peak position	Stress profile
Sharp interface	High	Interface	Highly localized
Linear gradient	Reduced	Tendon side	Moderately smooth
Exponential gradient	Reduced	Bone side	Asymmetric
Power-law ($n = 0.5$)	Strongly reduced	Shifted	Irregular
Power-law ($n = 2$)	Moderately reduced	Stable	Balanced

These results indicate that peak reduction alone is insufficient for model selection; spatial regularity and symmetry of the stress field are equally critical.

3.5. Power-Law Exponent Sensitivity Analysis

A sensitivity analysis was performed on the power-law exponent ($n = 1.5, 2, 3, 5$) using the refined 16-layer discretization.

Increasing n progressively shifts the stiffness transition toward the bone side, resulting in a broader redistribution of the stress field. Lower values of n maintain relatively high stress concentrations near the interface, whereas higher values reduce peak intensity but may introduce irregularities.

For $n = 5$, the stress profile becomes less regular, with a noticeable shift of the peak toward the bone region and the emergence of localized oscillations.

The configuration with $n = 3$ provides the optimal compromise between peak reduction and spatial regularity, and was therefore selected for subsequent analyses.

3.6. Refined Analysis: M17 vs. M18

The influence of Poisson's ratio was evaluated by comparing two configurations: M17 (constant ν) and M18 (variable ν), both based on the power-law model with $n = 3$ and 16 layers.

3.6.1. Stress Distribution

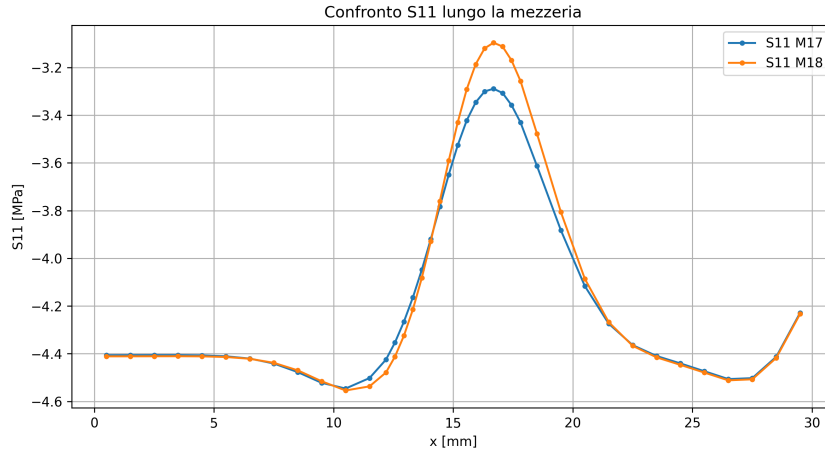


Figure 5: S11 distribution for M17 and M18. The two profiles are nearly coincident.

The stress distributions of M17 and M18 are nearly identical. Differences in peak magnitude and profile shape are negligible, indicating that the longitudinal stress field is primarily governed by the elastic modulus gradient.

3.6.2. Elastic Energy Distribution (SENER)

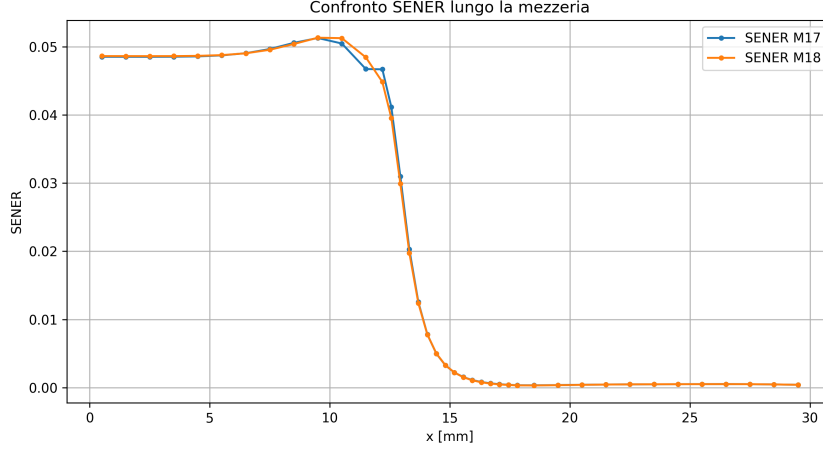


Figure 6: Strain energy density (SENER) along the central axis for M17 and M18.

Both models exhibit a concentration of elastic energy within the enthesis region, followed by a sharp decrease toward the bone.

The SENR profiles are almost perfectly overlapping, with differences below 0.1%.

3.6.3. Stress–Energy Relationship

A strong spatial correlation is observed between the stress gradient and the energy density distribution. Maximum energy accumulation occurs in regions of high stress variation, particularly near the tendon-side boundary of the enthesis, rather than at the location of the absolute stress peak.

This indicates that energy storage is governed by the combined stress–strain field and is highly sensitive to local stiffness gradients.

3.6.4. Quantitative Energy Metrics

Total elastic energy and peak SENR values are virtually identical for M17 and M18 (differences < 0.1%). A marginal smoothing of the energy gradient is observed in M18, without significant impact on the global mechanical response.

These results confirm that the elastic modulus gradient is the dominant factor controlling both stress redistribution and energy storage.

4. Discussion

4.1. Role of the Material Transition Law

The results demonstrate that the functional form of the material transition is the primary factor governing stress redistribution at the tendon–bone interface.

The sharp interface produces a highly localized stress concentration due to the abrupt stiffness mismatch (approximately 100:1), resulting in a concentrated peak in both stress and its spatial derivatives. In contrast, graded configurations distribute the load transfer over the enthesis region, significantly reducing stress localization.

However, the effectiveness of this redistribution strongly depends on the chosen transition law. Linear and exponential gradients introduce systematic asymmetry in the stress field, shifting the peak toward the tendon or bone side, respectively. This indicates that simple monotonic transitions are insufficient to ensure mechanically balanced load transfer.

The power-law formulation provides a more flexible framework, as the exponent n directly controls the spatial distribution of stiffness. The sensitivity analysis reveals a non-monotonic relationship between n and mechanical performance. While increasing n reduces peak stress and broadens the transfer region, excessively high values (e.g., $n = 5$) induce stiffness concentration near the bone, leading to secondary stress localization.

The optimal configuration ($n = 3$) represents a balance between stress reduction and spatial regularity, suggesting that effective graded interfaces require not only continuity but also controlled curvature of the stiffness profile.

4.2. Stress–Energy Relationship and Mechanical Implications

The analysis of strain energy density provides additional insight into the mechanical behavior of graded interfaces.

Maximum energy accumulation is observed in regions of high stress gradient rather than at the location of the absolute stress peak. This indicates that energy storage is governed by the combined stress–strain field and is strongly influenced by local stiffness variation.

In the sharp interface, energy is concentrated within a very limited volume, which may promote damage initiation due to localized mechanical overload. In contrast, graded configurations distribute energy over a larger region,

effectively reducing local energy density and enhancing mechanical robustness.

This behavior supports the interpretation of the enthesis as a functional transition zone that mitigates mechanical incompatibility between dissimilar tissues [2, 7]. More broadly, it highlights the importance of energy-based metrics as complementary design criteria for functionally graded materials.

4.3. Influence of Poisson’s Ratio

The comparison between M17 and M18 shows that spatial variation of Poisson’s ratio has a negligible effect on both stress and energy distributions (differences $< 0.1\%$).

Within the linear elastic, plane stress framework considered here, the mechanical response is dominated by the longitudinal stiffness gradient. Transverse deformation effects remain secondary under uniaxial loading conditions.

From a design perspective, this result is particularly relevant: it suggests that accurate control of the elastic modulus distribution is sufficient to capture the essential mechanics of graded interfaces, while variations in Poisson’s ratio can be neglected without significant loss of accuracy in this context.

4.4. Consistency with FGM Theory and Literature

The observed reduction in stress concentration with graded transitions is consistent with classical functionally graded material (FGM) theory [8].

Furthermore, the superior performance of non-linear gradients over linear ones aligns with previous studies on graded biomaterials and engineered interfaces [4, 3]. The present results extend these findings by providing a systematic comparison of transition laws and identifying an optimal exponent within a controlled parametric framework.

4.5. Limitations and Future Work

The present model includes several simplifying assumptions. The two-dimensional plane stress formulation neglects three-dimensional effects and out-of-plane stresses. Material behavior was assumed linear elastic and isotropic, whereas biological tissues exhibit anisotropy, viscoelasticity, and nonlinear response. The graded region was approximated through layer-wise discretization, and only static loading conditions were considered.

These simplifications were intentionally adopted to isolate the role of the material gradient. Future work should extend the analysis to three-dimensional geometries, incorporate more realistic constitutive models (e.g.,

fiber-reinforced or hyperelastic formulations), and investigate cyclic loading conditions to assess fatigue behavior and long-term stability of graded interfaces.

5. Conclusions

This study investigated the mechanical behavior of the tendon–bone interface through a systematic finite element analysis of functionally graded material transitions.

The results demonstrate that graded interfaces are highly effective in mitigating stress concentrations induced by abrupt stiffness mismatches. Among the investigated transition laws, the power-law distribution provides the most effective control of stress redistribution, achieving both peak reduction and spatial regularity of the stress field.

A non-monotonic relationship between the power-law exponent and mechanical performance was identified. The configuration with $n = 3$ represents an optimal balance between stress reduction and smooth load transfer, whereas higher exponents lead to stiffness localization and secondary stress concentration near the bone region.

The analysis of strain energy density further revealed that energy accumulation is governed by stiffness gradients rather than peak stress values. Graded configurations promote a distributed energy absorption mechanism across the enthesis, in contrast to the localized energy concentration observed in sharp interfaces.

The influence of Poisson’s ratio was found to be negligible under the considered loading conditions, indicating that the elastic modulus gradient is the dominant parameter controlling the mechanical response.

Overall, these findings provide quantitative insight into the design of biomimetic graded interfaces, highlighting the importance of controlling not only material continuity but also the spatial distribution of stiffness to achieve mechanically efficient load transfer.

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